

Stephen Hwang

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Technical skills

Certifications	AWS Certified Solutions Architect, MongoDB Database Administrator, MongoDB Developer
Programming languages	Python, MQL, SQL, R, JavaScript, Perl
Databases	MongoDB, Redis, SQL, Postgres, Pinecone, Qdrant, Elasticsearch, Redshift, BigQuery
Data engineering	Airflow, Kafka, Kinesis, Glue, Pub/Sub, SNS/SQS, Dataproc/Spark
AI tooling & frameworks	Spec-driven development with agent harnesses, Bedrock/Foundry/Gemini/HuggingFace, LangChain
Data sharing & visualization	FastAPI, Flask, Google Looker, Power BI, Tableau, ggplot, matplotlib
DevOps & infrastructure	Git, Subversion, Vercel, Docker, Kubernetes, Terraform, CloudFormation, Prometheus

Professional Experience

Partner Solutions Engineer	DataHub, San Francisco, CA	May 2026 - present
- Advised in creating a context layer with cross-platform column-level lineage, observability, and governance to support agentic workloads - Served as the primary technical point of contact for strategic technology partners and systems integrators, supporting joint go-to-market motions - Designed and validated system architectures that integrated DataHub with partner platforms (e.g., AWS, Google Cloud, Snowflake, Databricks) - Supported partner-sourced and co-sell opportunities through technical enablement, workshops, and live demos		
Senior Solutions Engineer, Applied AI	Research Triangle Institute, RTP, NC	May 2023 - May 2026
- Designed and prototyped new agentic approaches within FIPS-compliant and highly-regulated environments, integrating CI/CD pipelines and monitoring to automate the development lifecycle while ensuring 100% adherence to rigorous security and data sovereignty standards - Deployed advanced LLM-powered agents featuring persistent memory management and strict security protocols, ensuring high-context conversational continuity without compromising data privacy - Engineered a high-throughput backend using FastAPI and distributed databases, successfully scaling to 3M+ daily transactions with sub-100ms latency to provide real-time AI analytics for enterprise clients - Boosted search relevance by 80% by architecting a hybrid retrieval system, integrating full-text Lucene search with dense vector embeddings across sharded, distributed environments, to drastically improve user experience - Reduced client API integration timelines by 50% by designing a modular component library, standardizing deployment workflows, and accelerating the delivery of bespoke AI prototypes		
Solutions Architect, Enterprise Presales	MongoDB, New York, NY	Jan 2022 - Jul 2023
- Generated \$1.7M in net new consumption for enterprise clients in healthcare, financial services, and agriculture manufacturing - Designed a geolocation-based, sharded database deployment to support 20M weekly transactions with sub-50ms response times - Reduced proof-of-concept implementation times by 60% by creating boilerplate Python code artifacts for external enterprise teams - Designed event-driven solutions to support digital payments, consumer lending, and biometric data processing - Advised customers on best practices in data modeling, distributed computing, and event-driven architectures		
Senior Data Platform Engineer, Cloud Computing	Research Triangle Institute, RTP, NC	Jun 2015 - Jan 2022
- Reduced data latency by 80% by designing and deploying an event-driven ELT workflow that leveraged streaming (Kafka) and batch (Airflow) processing, improving overall stakeholder satisfaction and accelerating - Led cross-institutional teams of analysts and software developers as technical lead and project manager - Facilitated discussions with non-technical stakeholders to translate technical needs into actionable business goals		
Lead Backend Engineer	Oncotect, Raleigh, NC	Jan 2021 - Jan 2022
Designed an event-driven, distributed pipeline to aggregate and prepare clinical data for stakeholder consumption		
Backend Engineer	Johns Hopkins Medical Institutions, Baltimore, MD	Oct 2011 - Jun 2015
Served as data engineering lead on Stand Up 2 Cancer Epigenetics Dream Team (https://www.su2c.org/)		

Education

Johns Hopkins University , Baltimore, MD	M.S. Bioinformatics, <i>summa cum laude</i>	Aug 2013
Loyola University , Chicago, IL	B.S. Bioinformatics, <i>magna cum laude</i>	May 2011